Wumpus World Problem

4·1 Stench	4·2	4·3 Breeze	4·4 Pit
3·1	Breeze Stench gold	3·3 Pit	3·4 Breeze
2·1 Stench	2·2	2·3 Breeze	2·4
1·1	1·2 Breeze	1·3 Pit	14 Breeze

Wumpus → Stench

Pit → Breeze

PEAS

1. Performance Measure
   a) +1000 → agent comes out with gold
   b) - 1 → every action
   c) -10 → using arrow
   d) -1000 → pit / wumpus
   e) Game ends : Agent = dead or climbs out of cave

② Environment

A 4x4 grid of rooms
↳ Agent = [1,1] Right facing
↳ locn. of gold + Wumpus = random
↳ each square can start with pit P=0.2

③ Actuators
↳ motor: turn 90° left / right
↳ wheels : move forward
↳ Robotic arm : grab
↳ Robotic mech to shoot arrow

④ Sensors
a) Odor sensor — Stench (adj of wumpus)
b) Breeze perception
c) Camera - locate gold
d) Audio - Wumpus → killed → scream
e) Obstacle detection — Bump

Task env :

1. Partially observable - state is not known fully - wumpus, pit, gold
2. Deterministic - outcomes are known
3. Discrete - finite states, finite outcomes & moves
4. Sequential - order of moves matter
5. Static - world isn't evolving
6. Single Agent

Moves :

Every Some $\forall x$ ($\exists x$)
No $\sim\exists x$

First Order Logic

Q. All that glitter is not gold

$\sim[\forall x \quad glitter(x) \rightarrow gold(x)]$

$\Rightarrow \sim[\forall x \quad \sim glitter(x) \vee Gold(x)]$

$\Rightarrow \exists x \quad Glitter(x) \wedge \sim Gold(x)$

Q. All birds fly

$\forall x : birds(x) \rightarrow fly(x)$

Q. Some boys play cricket

$\exists x : boy(x) \rightarrow plays(x, cricket)$

Q. <u>Every</u> student loves <u>some</u> student

$\forall x \, (student(x) \rightarrow \exists y : (student(y) \wedge loves(x,y)))$

Q. Bill either takes analysis or geometr (but not both)

A ⇒ Bill takes analysis   takes(Bill, analy)
B ⇒ Bill takes geom.   takes (Bill, geom)

$(A \vee B) \wedge \sim(A \wedge B)$

Q. No student loves Bill

$\forall x: \text{student}(x) \to \sim\text{loves}(x, \text{Bill})$ or
$\sim \exists x: (\text{student}(x) \wedge \text{loves}(\text{Bill}))$

Q. Bill has atleast one sister

$\exists x: \text{sisterof}(x, \text{Bill})$

Q. Bill has no sister

$\sim \exists x: \text{sisterof}(x, \text{Bill})$

↑

there exist some $\sim$ ⇒ no one

Q. Lucy is a proff.

isproff (Lucy)

Q. All proff are people

$\forall x: (\text{isproff}(x) \rightarrow \text{people}(x))$

Q. John is the dean

isdean (John)

Q. Deans are proff.

$\forall x: (\text{isdean}(x) \rightarrow \text{proff}(x))$

Q. All proff consider their dean as friend or dont know.

$\forall x (\forall y (\text{isprof}(x) \wedge \text{isdean}(y) \rightarrow \text{friend}(x,y) \vee \sim \text{know}(x,y)))$

Q. Everyone is a friend of someone.

$\forall x \, \exists y : \text{friend}(x,y)$

Q. People critize only those who are not their friend

$\forall x (\forall y (\text{person}(x) \wedge \text{person}(y) \wedge \text{critize}(x,y) \rightarrow \sim \text{friend}(y,x)))$

Q. is John no friend of lucy?

is friend (John, Lucy)

Q Whoever can read is literate

$\forall x :$ can_read $(x) \rightarrow$ is literate (x)

$\sim$ read $(x) \vee$ literate (x)

Q. Dolphine are not Literate

$\forall x :$ dolphin $(x) \rightarrow \sim$ literate (x)

$\sim$ dolphin $(x) \vee$ literate $(x$

Q. Some dolphins are intelligent

$\exists x : [$ dolphin $(x) \wedge$ intelligent $(x)]$

Q. Some one who is intelligent cannot read

$\exists x : [$ intelligent $(x) \wedge \sim$ read $(x)]$

$\sim \Rightarrow \forall x$ ($\sim$ intelligent $(x) \vee$ read (x))

Q. Marcus was a man

→ man (Marcus)

Q. Marcus was a Pompeian

→ pompeian (Marcus)

Q. All pompeians are Romans

→ $\forall x$: pompeian(x) → roman(x)

Q. Caesar was a ruler

→ ruler (Caesar)

Q. All Romans were either loyal to Caesar or hated him

$\forall x$: roman(x) ⟶ [loyal(x, Caesar) ∨ hate(x, Caesar)]

generic

Q. Everyone is loyal to someone

$\forall x \exists y$: loyal(x, y)

Q. People only try to assassinate rulers they are not loyal to

$\forall x \forall y$: people $(x) \wedge$ ruler $(y) \wedge$ assassinates $(x, y) \rightarrow \sim$ loyal (x, y)

Q- Marcus tried to assasinate Caesar

tryassn. (Marcus, Caesar)

Q. All men are people

$\forall x \cdot$ man $(x) \rightarrow$ person (x)

Q. Gorilla is Black

$$gorilla(x) \rightarrow black(x)$$

Q. All kings are persons

$$\forall x : king(x) \rightarrow person(x)$$

Q. Every city in Maharashtra has a timple

$$\forall x : city(x) \wedge inMaha(x) \longrightarrow hastemple(x)$$

Q. An apple a day keeps. doctor away

$$\forall x \forall y ((Person(x) \wedge Apple(y) \wedge eats(x, y)) \rightarrow \sim need\,doctor(x))$$

Q. Anything anyone eats and is not killed by is food

$$\forall x \forall y : person(x) \wedge eats(x, y) \wedge \sim killed(x)$$

→ food(y)

Q. Square of 3 is 9

squareof(9, 3)

Resolution —

1. convert facts → FOL
2. FOL to CNF
3. Negate result
4. Draw resolution graph

Q.

a. John likes all kind of food

$$\forall x : food(x) \rightarrow likes(John, x)$$

b. Apple and vegetable are food

$$food(Apple) \wedge food(vegetable)$$

c. Anything anyone eats nd is not killed by is food

$$\forall x \forall y : \cancel{person(x) \wedge} \; eats(x,y) \wedge \neg killed(x) \rightarrow food(y)$$

d- Anil eats peanuts and still alive

$$eats(Anil, peanuts) \wedge alive(Anil)$$

e - Harry eats everything that Anil eats

$$\forall x : eats(Anil, x) \rightarrow eats(Harry, x)$$

Steve only likes easy courses

↳ $\forall x: easy(x) \rightarrow likes(Steve, x)$ ✗

① ↳ $\sim easy(x) \vee likes(steve, x)$

Science courses are hard

↳ $\forall x: science(x) \rightarrow \sim easy(x)$

② ↳ $\sim science(x) \vee \sim easy(x)$

All course in the basket_weaving dept are easy

↳ $\forall x: inbasket\_weaving(x) \rightarrow easy(x)$

③ ↳ $\sim inbw(x) \vee easy(x)$ ✗

BK301 is a basket_weaving course

↳ in basket_weaving (BK301)

④ inbw (BK301) ✗

⇒ what course does steve like?

$likes(steve, x)$

↳ $\sim likes(steve, x)$

$\sim likes(\cancel{steve}, x)$ $\qquad$ $\sim easy(x) \vee likes(\cancel{steve}, x)$

$\sim \cancel{easy(x)}$ $\qquad$ $\sim inbw(x) \vee \cancel{easy(x)}$

$\sim \cancel{inbw(x)}$ $\qquad$ $\cancel{inbw(BK301)}$

$x/BK301$

null

$\therefore\ x = \underline{\underline{BK301}}$

1 Q. Marcus was a man

$\rightarrow$ man (Marcus)

2 Q. Marcus was a Pomp

$\rightarrow$ pompeian (Marcus)

3 Q. All pompeians are Romans

$\rightarrow \forall x :$ pompeian$(x) \rightarrow$ roman(x)

$\sim$pomp$(x) \vee$ roman(x)

4 Q. Caesar was a ruler

$\rightarrow$ ruler (Caesar)

5 Q. All Romans were either loyal to Caesar or hated him

$\forall x :$ roman$(x) \longrightarrow$ [loyal $(x$, Caesar$) \vee$ hate $(x$, Caesar$)$]

$\sim$roman$(x) \vee$ loyal$(x, c) \vee$ hate(x, c)

6 . Everyone is loyal to someone

generic

$\forall x \exists y :$ loyal (x, y)

loyal (x, f(x))

7 Q. People only try to assassinate rulers
are not loyal to

$\forall x\ \forall y$: people (x) $\wedge$ ruler (y) $\wedge$
assassinates (x, y) $\rightarrow$
$\sim$ loyal (x)

$\sim$ people (x) $\vee$ $\sim$ ruler (y) $\vee$
$\sim$ ass. (x, y) $\vee$ $\sim$ loyal (x)

8 Marcus tried to assassinate Caesar
tryassn. (Marcus, Caesar)

9 Q. All men are people
$\forall x$. man (x) $\rightarrow$ person (x)
$\sim$ man (x) $\vee$ person (x

$\Rightarrow$ $\sim$ hate (Marcus, Caesar)

$\cancel{\sim hate(M, C)}$ $\quad \sim roman(\overset{m}{\cancel{x}}) \lor loyal(\overset{m}{\cancel{x}}, c) \lor \cancel{hate(x, c)}$ m

marcus/x

$\cancel{\sim roman(m)} \lor loyal(m, c) \qquad \sim pomp(\overset{m}{\cancel{x}}) \lor \cancel{roman(\overset{m}{x})}$

marcus/x

$loyal(m, c) \lor \cancel{\sim pom(m)} \qquad \cancel{pomp(x)}$

m/x

$\cancel{loyal(m, c)} \qquad \sim people(x) \lor \sim ruler(y) \lor \sim ass(x, y) \lor \cancel{\sim loyal(x)}$

$\sim people(m) \lor \sim ruler(c) \lor \cancel{\sim ass(m, c)} \qquad \cancel{ass(m, c)}$

$\sim people(m) \lor \cancel{\sim ruler(c)} \qquad \cancel{ruler(c)}$

~~~people~~(m)
person

$\sim man(x) \vee \cancel{person(x)}$

$\sim man(x)$

$man(m)$

null

$\therefore$ true
~~~

① All people who are earning are happy

$\forall x : earning(x) \rightarrow happy(x)$

$\Rightarrow \sim earning(x) \vee happy(x)$

② All happy people smile

$\forall x : happy(x) \rightarrow smile(x)$

$\Rightarrow \sim happy(x) \vee smile(x)$

③ Someone is earning

$\exists x : earning(x)$

$\Rightarrow earning(E(a))$

$\sim \exists x : smiling(x)$

$\forall x : \sim smiling(x) = \sim smiling(x)$

$\cancel{\sim smiling(x)}$ $\qquad$ $\sim happy(x) \vee \cancel{smiling(x)}$

$\sim \cancel{happy(x)}$ $\qquad$ $\sim earning(x) \vee \cancel{happy(x)}$

$\cancel{\sim earning(x)}$ $\qquad$ $\cancel{earning(E(a))}$

$E(a)/x$

null

$\therefore$ proved.

Forward Chaining

↳ based on data availablity → decision taken

Q. 1) It is crime for an American to sell weapons to the enemy of America

$$American(x) \wedge weapon(y) \wedge enemy(z, America) \wedge sells(x, y, z) \longrightarrow Criminal(x)$$

x

2) Country nano is an enemy of America

$$enemy(Nano, America)$$

sells x fact

3) Nano has some missiles

$$\exists x : missiles(x) \wedge owns(nano, x)$$

4) All of its missiles were sold to it by Col. West

$$\forall x : missiles(x) \wedge owns(Nano, x) \rightarrow$$

sold (Col. West, x, Nano)

5) Col. West who is an American

American (Col. West)

6 missiles (x) $\rightarrow$ weapon (x)

$\rightarrow$ list facts (LHS)

American (Col. West) Missile (x) own (Nono, x)

enemy (Nono, America)

\ Root → check if any statement's LHS present

Q.

1. Fred is a collie
collie (Fred)

2. Sam is Fred's master
master (Fred, Sam)

3. The day is Saturday
day (Saturday)

4. It is cold on Sat
cold (Sat.)

5 Fred is trained
trained (Fred)

6. Trained Collies are good dogs
$\forall x$: collie(x) ^ trained(x) → gooddog(x)

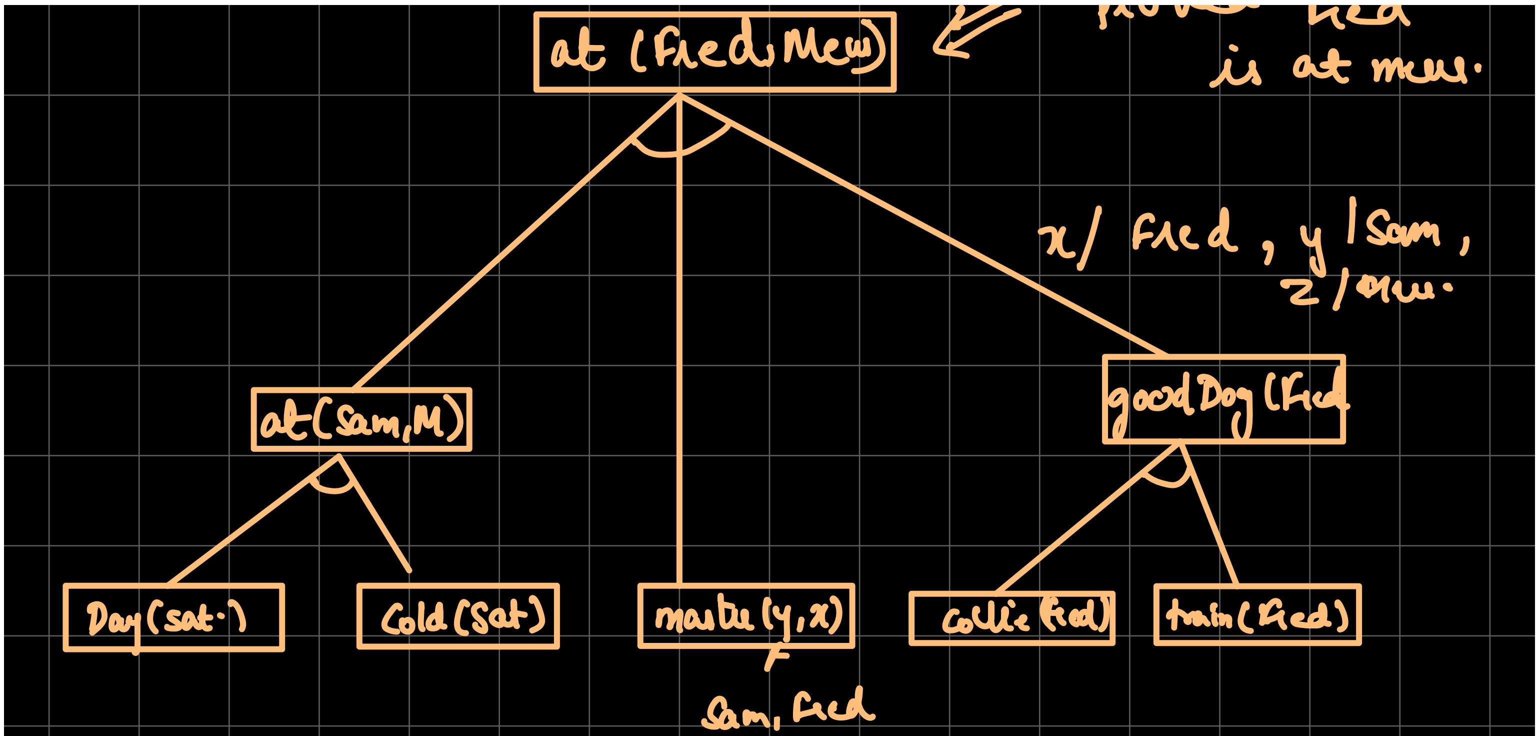
7. If a dog is good & has a master and at some place, then he will be with his master

$\forall x \forall y \forall z$: goodDog(x) ^ master(y, x) ^ at(y, z) → at(x, z)

8. If day is sat. & day is cold, then Sam is at museum

day(sat.) ^ cold(Sat) → at (sam, museum)

← proved Fred

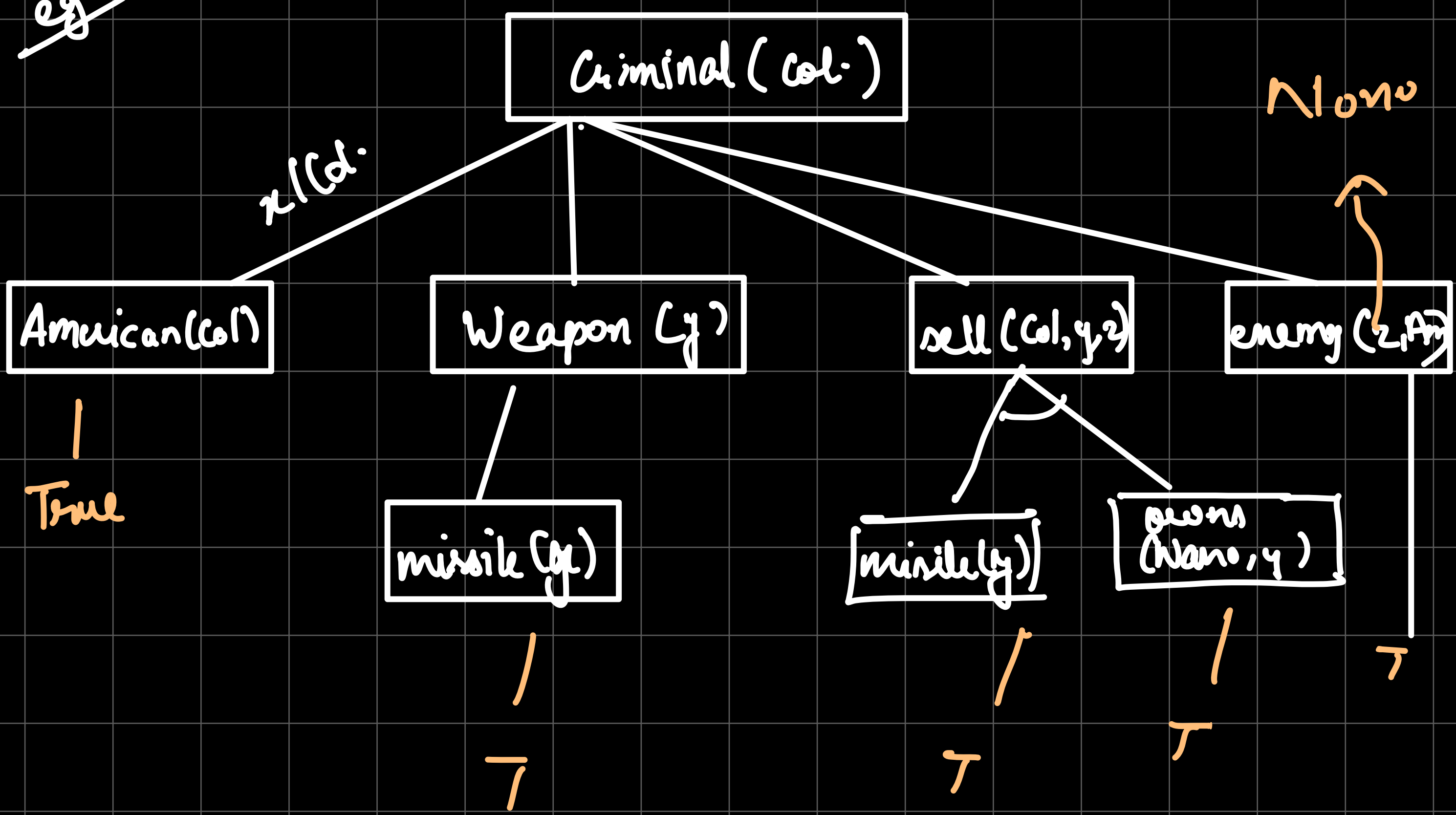


at (Fred, Mew)
is at mew.
x/ Fred, y/Sam, z/Mew.
at(Sam,M)
goodDog(Fred
Day(sat)
Cold(Sat)
master(y,x)
Sam, Fred
Collie(Fred)
train(Fred)

Backward Chaining

Goal → Facts

eg America



Q. 1. Every student smiles

2. No one talks

3. Atleast one student failed History

4. Every person who buys an insurance policy is smart

5. No person buys an expensive policy

1) $\forall x : student(x) \longrightarrow smiles(x)$

2) $\exists x : talks(x) \Rightarrow \sim \exists x \; talks(x)$

$= \forall x \sim talks(x)$

3) $\exists x : (student(x) \wedge fail(x, History))$

4) $\forall x : person(x) \wedge buys(x, insurance) \rightarrow smart(x)$

5) $\exists x : buys(x, exp \cdot policy) \Rightarrow \sim [\quad]$

$\forall x : \sim buys(x, exp \cdot policy)$

Q. a) Ravi likes all kind of food

b) Apple & Chicken are food

c) Anything anyone eats and is not killed is food.

d) Ajay eats peanuts & still alive.

e) Rita everything that Ajay eats.

b a) $\forall x: \text{food}(x) \rightarrow \text{likes}(\text{Ravi}, x)$

$\sim \text{food}(x) \vee \text{likes}(\text{Ravi}, x)$

b) $\text{food}(\text{Apple}) \wedge \text{food}(\text{Chicken})$

$\hookrightarrow$,

c) $\forall x \forall y: \text{eats}(x, y) \wedge \underline{\sim \text{killed}(x)} \rightarrow \text{food}(y)$

$\sim \text{eats}(x, y) \vee \text{killed}(x) \vee \text{food}(y)$

d) $\text{eats}(\text{Ajay}, \text{peanuts}) \wedge \underline{\text{alive}(\text{Ajay})}$

,

e) $\forall x: \text{eats}(\text{Ajay}, x) \rightarrow \text{eats}(\text{Rita}, x)$

$\sim \text{eats}(\text{Ajay}, x) \vee \text{eats}(\text{Rita}, x)$

alive(x) → ~killed(x)   ~killed(x) → alive(x)

~alive(x) ∨ ~killed(x)   killed(x) ∨ alive(x)

GOAL = Ravi likes Peanuts

↳ likes(Ravi, Peanuts)

Negat ~likes(Ravi, Peanuts)

~~~likes~~(Ravi, Pea.)   ~food(x) ∨ ~~likes~~(R, x)

x/P

~~~food~~(Pea)   ~eats(x, y) ∨ killed(x) ∨ ~~food~~(y)

~~~eats~~(x,y) ∨ killed(x)   ~~eats~~(A, P)

x/A
y/P

~~killed~~(Aj)   ~alive(x) ∨ ~~~killed~~(x)

x/Aj

~alive(Aj)   alive(Aj)

null   ∴ proved.
~~~

Bayesian Network

→ Real world appn. are probabilistic in nature & to represent reln. b/w multiple events ⇒ BNN

→ Bayesian reasoning determines probability of an event with uncertain knowledge

BNN :

→ DAG

→ Table of conditional probabilities

↓

Each node condn. prob $P(X_i \mid Parent(X_i))$

⇓

influence of parent

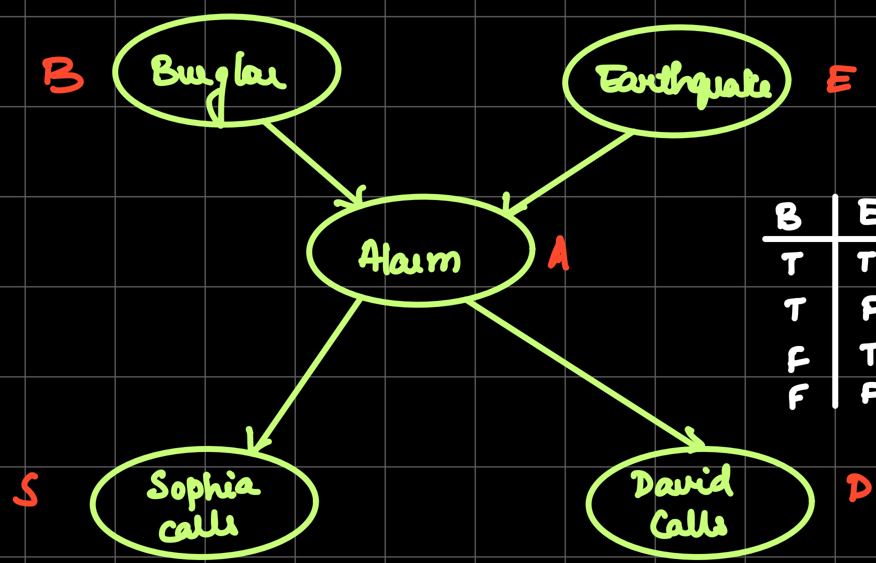
Joint Probability Distribution + Condn. Probability

Burglar Problem

↳ 2^k possiblities

T	0.002
F	0.998

T	0.001
F	0.999



P(A | B,E)

B	E	P(A) = T	= F
T	T	0.95	0.05
T	F	0.94	0.06
F	T	0.29	0.71
F	F	0.001	0.999

P(S | A)

Alarm	T	F
T	0.9	0.1
F	0.05	0.95

P(D | A)

Alarm	T	F
T	0.7	0.3
F	0.01	0.99

B & E are independent

A → B, E

D, S → A

$$P(B, E, A, S, D) = P(B) \cdot P(E) \cdot P(A \mid E, B) \cdot P(S \mid A) \cdot P(D \mid A)$$

$\Rightarrow$ alarm sounded $\rightarrow$ no B no E but S D ✓

$$P(\sim B, \sim E, A, S, D) =$$

$$P(\sim B) \cdot P(\sim E) \cdot P(A \mid \sim B \sim E) \cdot P(S \mid A) \cdot P(D \mid A)$$

from CPT,

$$= 0.999 \times 0.998 \cdot 0.001 \cdot 0.9 \cdot 0.7$$

$$= 0.000628$$

$\Rightarrow$ david calls

$$P(D) = P(D \mid A) \cdot P(A) + P(D \mid \sim A)\, P(\sim A)$$

$\rightarrow$ both ind so just multiply

$$P(A) = P(A|BE) \cdot P(BE) + P(A|\sim B \sim E)\, P(\sim B \sim E) +$$
$$P(A|B \sim E) \cdot P(B \sim E) + P(A|\sim B E)\, P(\sim B E)$$

$\Rightarrow P(B) \Rightarrow J \,\&\, M$ ✓

Now, if asked about the root

of reln. b/w nodes with no direct

connection the

Marginal Prob

$P(X|Y)$ $\dfrac{P(X,Y)}{P(Y)}$

Joint keep X Y const.

Q $P(A1) = 0.1$

$P(A2) = 0.2$

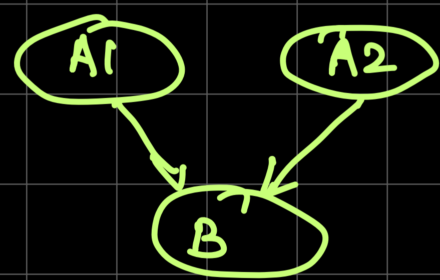
$P(B | A1, A2) = 0.8$

$P(B | A1, \sim A2) = 0.7$

$P(B | \sim A1\ A2) = 0.6$

$P(B | \sim A1\ \neg A2) = 0.5$

$P(A2 | B, A1)$



$P(B, A1, A2) =$

$$P(A_1)\ P(A_2) \cdot P(B | A_1 A_2)$$

$$= 0.1 \times 0.2 \times 0.8 = 0.016$$

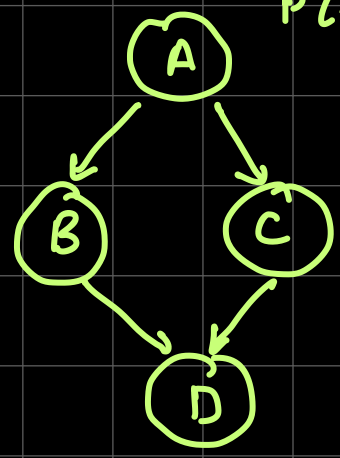
$$\Rightarrow \quad P(A2 | B A1) = \frac{P(A1, A_2, B)}{P(B A_1)} = 0.22$$

$$P(B, A_1) = P(A_1, A_2, B) + P(A_1, \sim A_2, B)$$

$$= 0.016 + 0.056$$

$$= 0.072$$

$P(A) = 0.75$



$P(B|A) = 0.2$

$\sim A \quad 0.5$

$P(C|A) = 0.7$

$\sim A = 0.25$

$D|BC$	0.3	$\sim B\, C$	0.1
$D|B\sim C$	0.25	$\sim B \sim C$	0.35

$$P(D|A) = \frac{P(D,A)}{P(A)}$$

$$P(A,B,C,D) = P(A) \cdot P(B|A) \cdot P(C|A) \\ P(D,BC)$$

$$P(D,A) = P(A,B,C,D) + P(A,\sim B,C,D) \\ + P(A,B,\sim C,D) + P(A,\sim B,\sim C,D)$$